

Measuring and Managing Risk at Scale: Index Insurance and Aflatoxin Early-Warning

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Agenda



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| ① Motivation & Context | ⑤ Prediction Overview |
| ② Questions & Contributions | ⑥ Evaluation Plan & Metrics |
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| ④ Our Method | ⑧ Implications & Takeaways |

Our design delivers higher welfare and improves further with joint (multi-zone) optimization

- **Single-zone (per-zone) design:** $RIB = 0.524$ (VMX) vs. -0.163 (Chantararat) and -1.909 (NN).
- **Designing jointly (portfolio/MZ):** RIB rises from $0.524 \rightarrow 0.592$.
- Premiums are *risk-adjusted*: priced based on mean payout and portfolio $CVaR_{1-\epsilon}$.
- Contracts remain **simple and interpretable** (piecewise-linear with cap), aiding policy adoption.

Policy-relevant takeaway

Multi-zone design uses diversification and capital reallocation to make contracts more welfare improving.

¹ RIB^* benchmarks the utility gain against perfect insurance: $RIB^* := \frac{CE(W^{ins}) - CE(W^0)}{CE(W^{PI}) - CE(W^0)}$.

Costly verification + covariate shocks make indemnity hard to scale

- Farmers face a lot of risk, and the lack of risk management tools forces them to use coping strategies that hurt their long term welfare.
- Traditional insurance is prohibitively costly in most developing countries due to lack of data and high verification costs.
- Moral hazard, adverse selection, and the presence of large covariate shocks make the problem of agricultural insurance especially hard.

Index insurance reduces verification & moral hazard — but real-world performance hinges on design & pricing

- **What it is.** Payouts are tied to an easily observable and measureable index (e.g., rainfall) rather than manual loss verification.
- **Why it helps.** Automating payouts reduces loss verification costs and prevents moral hazard; contracts are easy to scale.
- **Problems.** Programs operate at scale in India, Mexico, Kenya, Malawi, Tanzania, and other countries, but demand is often low and products usually need subsidizing.
- **Design challenge.** Farmers are very price sensitive, so we need to make insurance cost-effective:
 - ▶ simple, deductible-like piecewise-linear contracts (interpretable, adjustable);
 - ▶ realistic, risk-adjusted premium, e.g. $\pi = \mathbb{E}[I] + c_\kappa K(I)$;
 - ▶ portfolio (multi-zone) design so capital is computed on entire portfolio: $\sum_z s_z I_z$.

II is typically subsidized, often bundled with credit, delivered via PPPs

- Index insurance is a named-peril insurance and it is often tied to access to credit.
- Offered at different levels (micro level, meso level, and macro level)
- Often based on public-private partnerships. Research institutions usually design the product, and insurance companies (or governments) sell it.

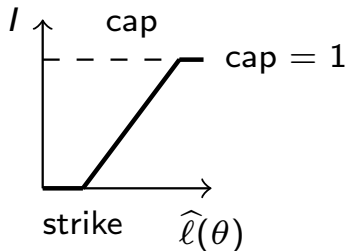
Piecewise-linear contracts: understandable and adjustable

Contract form

$$I(\theta) = \min\{\max[0, a\hat{\ell}(\theta) - b], 1\}$$

where $\hat{\ell}(\theta)$ is predicted loss, $a \geq 0$ is the slope, $b \geq 0$ is the strike/intercept, and payouts are capped at 1 (normalized).

- *Deductible-like*: theoretically optimal (Arrow (1974)).
- *Interpretable*: simple contracts are easier to explain to farmers, this is important for take up (Ahmed, McIntosh, and Sarris (2020)).
- *Popular*: this contract structure is the most commonly used by practitioners (Gamage (2021)).



0 below strike; slope a until cap; flat thereafter.



Questions and Contributions

What we ask and what we contribute

Research Questions

- How can we **design simple (piecewise-linear)** index insurance contracts that maximize farmer welfare under **risk-adjusted pricing**?
- What do we gain by **simultaneously designing contracts** across zones when capital is priced on the portfolio?

Contributions

- A **utility-maximizing** design with premiums tied to $\text{CVaR}_{1-\epsilon}$.
- A **multi-zone portfolio** program with and a simple proof that it improves over independent zone design.

Index Insurance Literature

- **Impact of Index Insurance:** Overall, there is evidence that index insurance reduces reliance on detrimental risk-coping strategies, increases investment, and leads to riskier, but more profitable production decisions (Jensen and Barrett (2017); Cole et al. (2013); Mobarak and Rosenzweig (2013); Karlan et al. (2014)).
- **Demand for Index Insurance:** Demand for index insurance tends to be low and highly price sensitive (Jensen and Barrett (2017); Cole et al. (2013); Cai, De Janvry, and Sadoulet (2020), Casaburi and Willis (2018)).
- **Design of Index Insurance:** There has been relatively little research studying the design of index insurance. The method developed by Chantarat et al. (2013) is the most commonly used in academic publications (Jensen, Stoeffler, et al. (2019); Flatnes, Carter, and Mercovich (2018)). More recently, Chen et al. (2023) used a NN approach for contract design.



Methods Overview

Baseline Method: Chantarat et al. (2013)

Developed by Chantarat et al. (2013), used in Kenya Index Based Livestock Insurance (IBLI) program. Consist of two steps:

- 1 Train model to predict loss
- 2 Choose strike value, b , that maximizes correlation between payouts and losses

Pros: simple, interpretable

Cons: can't incorporate constraints, doesn't take into account premium

NN Approach: Chen et al. (2023)

Train neural network to solve the following optimization problem:

$$\begin{aligned} \max \quad & \mathbb{E}[u(w_0 + 1 - \ell + I - \pi)] \\ \text{s.t.} \quad & \pi = \lambda \mathbb{E}[I] \\ & \pi \leq \bar{\pi} \end{aligned}$$

Use CARA utility function:

$$\begin{aligned} u(w) &= \frac{-1}{\alpha} e^{-\alpha w} \\ \alpha &> 0 \end{aligned}$$

Pros: Directly maximizes utility, admits price constraints.

Cons: Not interpretable. hard to adjust, large data requirements

We predict losses, then optimize simple contracts under capital-aware pricing

Our approach is most similar to the baseline method. We predict-then-design. The steps are:

- 1 Train a traditional ML model to predict loss
- 2 Use model predictions and ground truth to design contracts

This yields simple and interpretable contracts.

Method comparison: objective, pricing, contract class, interpretability

	Chantarat	Chen (NN)	Ours (VMX)
Objective	Maximize correlation of payouts with losses	Maximize expected utility	Maximize expected utility
Pricing	Not incorporated	$\pi = \lambda \mathbb{E}[I]$	$\pi = \mathbb{E}[I] + c_{\kappa}(\text{CVaR}_{1-\varepsilon}(I) - \mathbb{E}[I])$
Contract class	Piecewise-linear	Non-parametric	Piecewise-linear with cap
Interpretability	High	Low	High

Method comparison: other features

	Baseline	Chen	VMX
Simple contracts	✓	✗	✓
Price constraints	✗	✓	✓
Payout frequency constraints	✗	✗	✓
Different premiums defn's	✗	✓	✓
Multi-zone contract design	✗	✗	✓



Our Method

CVaR: definition, coherence, and how it enters pricing

Definitions (for loss L , CDF F_L , confidence $\epsilon \in (0, 1)$)

$$\begin{aligned}\text{VaR}_{1-\epsilon}(L) &:= \inf\{\ell \in \mathbb{R} : F_L(\ell) \geq 1 - \epsilon\}, \\ \text{CVaR}_{1-\epsilon}(L) &:= \mathbb{E}[L \mid L \geq \text{VaR}_{1-\epsilon}(L)]\end{aligned}$$

Risk-adjusted premium (normalized payouts I)

$$\pi = \mathbb{E}[I] + c_\kappa \left(\text{CVaR}_{1-\epsilon}(I) - \mathbb{E}[I] \right)$$

Why CVaR?

- It is a coherent risk measure: monotonicity, translation invariance, positive homogeneity, **subadditivity**.
- CVaR is popular both in industry and academic works. Used for solvency requirements in Switzerland, Canada, and USA.

Premium definition from *Risk Modeling for Appraising Named Peril Index Insurance Products: A Guide for Practitioners* Mapfumo, Groenendaal, and Dugger (2017)

Premium Principles

A premium principle is a functional that maps an insurance contract to a price (premium). There are several premium principles that are common in the actuarial literature, and our method is compatible with most of them:

Description	Definition
Expected value	$(1 + \alpha)\mathbb{E}[I]$
Standard deviation	$\mathbb{E}[I] + \alpha\sigma(I)$
Variance	$\mathbb{E}[I] + \alpha\sigma^2(I)$
Exponential	$\frac{1}{\alpha} \log \mathbb{E}[e^{\alpha I}]$
Dutch	$\mathbb{E}[I] + \beta\mathbb{E}[(I - \alpha\mathbb{E}[I])_+]$

Idealized Model

$$\max_{a,b,\pi} \mathbb{E} [U(w_0 + 1 - \ell - \pi + I(\theta))] \quad (1)$$

$$\text{s.t.} \quad I(\theta) = \min \left\{ \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}, 1 \right\} \quad (2)$$

$$\pi = \mathbb{E} [I(\theta)] + c_{\kappa} [\text{CVaR}_{1-\epsilon_K} (I(\theta)) - \mathbb{E}[I(\theta)]] \quad (3)$$

$$\pi \leq \bar{\pi}. \quad (4)$$

Convex Relaxation

$$\max_{a,b,\pi} \mathbb{E} \left[U \left(w_0 + 1 - \ell - \pi + \underline{I(\theta)} \right) \right] \quad (5)$$

$$\text{s.t. } \overline{I(\theta)} = \max \left\{ 0, a\hat{\ell}(\theta) + b \right\}$$

$$\underline{I(\theta)} = \min \left\{ a\hat{\ell}(\theta) + b, 1 \right\}$$

$$\pi = (1 - c_\kappa) \mathbb{E} \left[\overline{I(\theta)} \right] + c_\kappa \text{CVaR}_{1-\epsilon_K} \left(\overline{I(\theta)} \right) \quad (6)$$

$$\pi \leq \overline{\pi}.$$

CP Formulation: Rockafellar and Uryasev (2002)

$$\begin{aligned} \max_{a,b,\{\gamma_K^j\},\{\alpha^j\},\{\omega^j\},t_K,K} \quad & \frac{1}{N} \sum_j U(w_0 + 1 - \ell^j - \pi + \omega^j) \\ \text{s.t.} \quad & \pi = \frac{(1 - c_K)}{N} \sum_j \alpha^j + c_K K \\ & t_K + \frac{1}{\epsilon_K} \sum_j p^j \gamma_K^j \leq K \\ & \gamma_K^j \geq \alpha^j - t_K, \forall j \\ & \gamma_K^j \geq 0, \forall j \\ & \alpha^j \geq a \hat{\ell}(\theta^j) + b, \forall j \\ & \alpha^j \geq 0, \forall j \\ & \omega^j \leq a \hat{\ell}(\theta^j) + b, \forall j \\ & \omega^j \leq 1, \forall j \\ & \pi \leq \bar{\pi}. \end{aligned}$$



Prediction Overview

Prediction: Problem Overview

- For each plot i , we have several time-series of different weather variables, sampled at potentially different frequencies:

$$\begin{bmatrix} x_{i,1} & \dots & x_{i,16} & \dots & x_{i,176} \\ y_{i,1} & y_{i,2} & \dots & y_{i,16} & \dots & y_{i,176} & \dots & y_{i,180} \\ z_{i,1} & z_{i,2} & \dots & z_{i,16} & \dots & z_{i,176} & \dots & z_{i,180} \end{bmatrix}$$

- These weather variables are rainfall, temperature, and evapotranspiration.

Feature Extraction

- Time-series classification (TSC) is different from standard problems because the data points are ordered, and the order matters. The space of potential features is also large, e.g. we can take 3-day dry spells, 5-day dry spells, number of high temperature days, etc.
- Ruiz et al. (2021) compared many different specialized TSC algorithms and they found that the RoCKET algorithm (Dempster, Petitjean, and Webb (2020)) generally performed best.
- We use RoCKET, Catch22 (Lubba et al. (2019)), and raw time-series for feature extraction.

Prediction overview — features, models, validation

- **Preprocessing:** uniform scaling, quantile transformer
- **Feature extraction:** Catch22, RoCKET, raw time-series
- **Prediction algorithm:** Random Forests (RF), Gradient Boosting Regressor (GBR), Elastic Net, LASSO



Evaluation Data, Plan, and Metrics

Data in Index Insurance Research

- One of the biggest barriers in AI research is the lack of granular yield data for LMIC's.
- Most previous studies either use a small dataset from a pilot index insurance program in Kenya (Chantararat et al. (2013); Flatnes, Carter, and Mercovich (2018); Estefania-Salazar et al. (2024)) or use data from a rich country (Chen et al. (2023)).
- I collaborated with the Bank of Thailand and obtained access to a novel dataset of losses from natural disasters in Thailand.

Thai agriculture is rain-fed and smallholder-dominated → high weather exposed

- Characterized by rural, smallholder farmers, with an average farm size of 5.7 acres. Average in sub-Saharan Africa, South Asia, and East Asia is ~ 4 acres.
- 76% of agriculture in Thailand is rain-fed. Globally, this ranges from 60% in South Asia to over 90% in sub-Saharan Africa.
- Consistent exposure to floods and droughts, especially in northeast region. LMIC's generally face more flood and drought risk than richer countries (Winsemius et al. (2018)).
- $\sim 80\%$ of poor households are employed in agriculture, and $\sim 30\%$ of overall Thai labor force is employed in agriculture

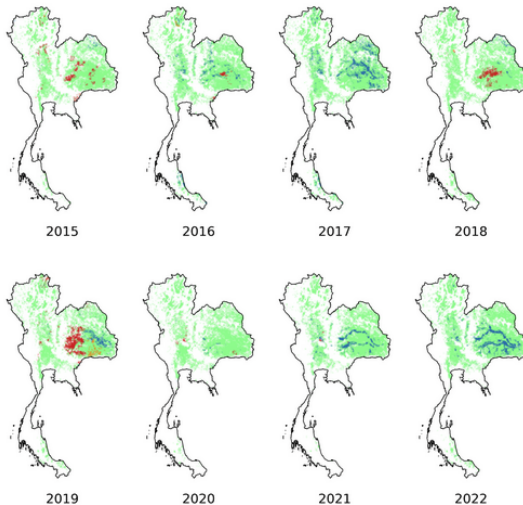
Thai Data: Administrative Data Sources

- **Thailand Department of Agricultural Extension (DOAE):** This dataset contains detailed information on the planting activities of over 3.1 million Thai farmers. Information includes plant date, plot size, rice variety, etc
- **Thai Government Disaster Relief Data:** This data is from the current national disaster relief program. It contains information about the disaster type, losses, and disaster date.

Remote Sensing Data Sources

- **Climate Hazards Center InfraRed Precipitation with Station data (CHIRPS):** daily rainfall data 1981-present
- **MOD21A1 Land Surface Temperature:** daily temperature data, available 2000-present
- **FLDAS Famine Early Warning Systems Network:** monthly evapotranspiration data, available 1992-present

Disaster Activity Map



Evaluation plan: leave-one-year-out (train on $y' \neq y$, test on y)

- We use leave one year out cross validation.
- In each CV fold, data from year y is used for testing. Data from $y' \neq y$ is used for insurance design (prediction model training and contract design).
- This gives us out-of-sample predictions for every observation in our data. We then compute utility-based performance metrics across all out-of-sample predictions.
- Out-of-sample predictions used for contract design and evaluation, pricing held constant across methods.

Evaluation Details

- We use the training data and the same method to compute the premium for all methods.
- We modify Chen's method to use the same premium definition as ours.
- We use simulated data to estimate the CVaR when calculating the premium.
- We use a CRRA utility function with $\alpha = 1.5$ (Flatnes, Carter, and Mercovich (2018); Estefania-Salazar et al. (2024); Jensen, Stoeffler, et al. (2019))

Metrics: RIB and behavior (precision/recall; good-gain vs bad-loss)

- We focus on the Relative Insurance Benefit (RIB) (Kenduiwo et al. (2021); Estefania-Salazar et al. (2024)) as our main metric.
- Compares a proposed insurance contract to perfect insurance. It measures the welfare gains from perfect insurance that the proposed contract recovers.
- Interpretation: if RIB of an insurance J is 50%, we can think of it as meaning that it is 50% as good as perfect insurance. In other words, it yields half as many welfare gains as perfect insurance.

Relative Insurance Benefit (RIB)

Let w_i^J be the wealth of farmer i under insurance contract J . We denote perfect insurance with PI and no insurance with NI

$$CE(J) : u^{-1}(\mathbb{E}[u(w^J)])$$

$$\Delta CE(J) : = CE(J) - CE(NI)$$

$$RIB(J) : \frac{\Delta CE(J)}{\Delta CE(PI)}$$

$$\text{Utility : } u(w) = \frac{w^{1-\alpha}}{1-\alpha}$$

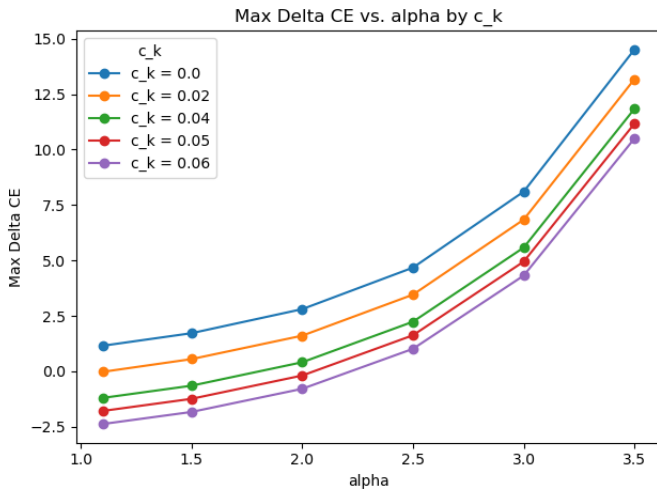


Results (SZ $\rightarrow$ MZ)

SZ: VMX achieves RIB 0.524 vs -0.163 (Chantararat) and -1.909 (NN)

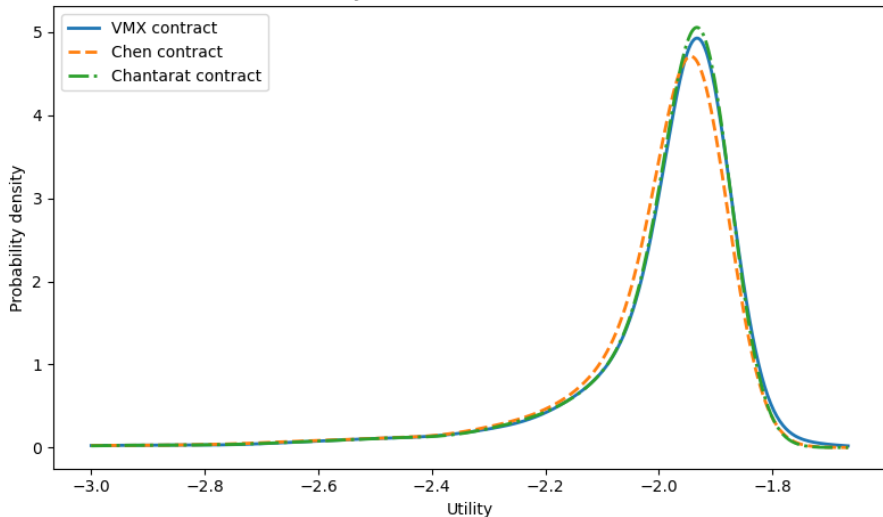
Method	RIB	Premium	Cost	Cost PI
VMX	0.524	0.014	0.017	0.027
Chantararat	-0.163	0.004	0.002	0.027
Chen	-1.909	0.074	0.062	0.027

Sensitivity: perfect insurance attractive only at low c_k ;
gains rise with risk aversion



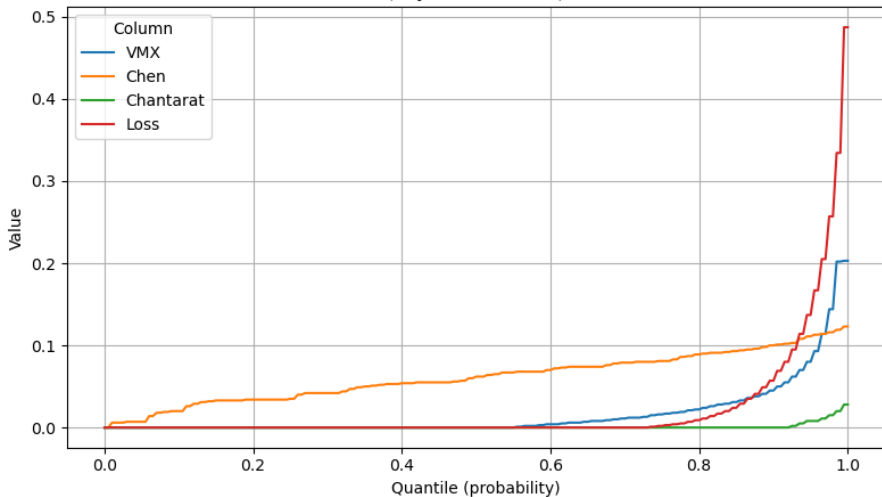
Outperforms in the entire distribution

Utility distributions: VMX vs. Chen



Payout and Loss Quantile Functions

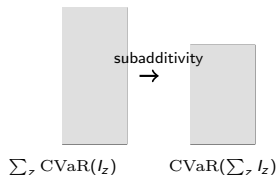
Empirical Quantile Function
(Payouts and Loss)



From SZ results → Why multi-zone?

What we just saw (SZ):

- VMX (SZ) improves welfare: RIB = 0.524 vs. Chantarat -0.163, Chen (NN) -1.909.
- Better behavior: more payouts at the tails, less unnecessary payouts.



But SZ optimizes per zone:

- Prices capital as if it depended on I_z alone; ignores cross-zone dependence.
- Ignores diversification benefits (CVaR is subadditive)
- Independent design can misallocate coverage across zones.

Capital charge under CVaR: portfolio $\leq$ sum of standalones.

Next (MZ): choose $\{a_z, b_z\}_z$ *jointly* and allocate capital $\{k_z\}_z$ with $\sum_z k_z = K$ to improve welfare gains per unit of *portfolio* risk.

Why multi-zone: Capital is priced on the portfolio

- **Capital is portfolio-based.**
Required capital depends on $S = \sum_z s_z I_z$; premiums depend on *portfolio* risk.
- **Why SZ is misspecified.** SZ treats capital as if it depended only on I_z , so program only uses zone's tail, not the portfolio tail.
- **Implication.** SZ ignores cross-zone dependence and yield the wrong FOCs.

Simple 2-zone case (intuition).

Optimize $\theta_1 \in \{a_1, b_1\}$ with I_2 fixed. Under a mean–**stdev** premium, the MZ FOC includes a **covariance** term with I_2 ; SZ does not.

2-zone mean–stdev premium

Let $S = s_1 I_1 + s_2 I_2$ and $\sigma_S = \sqrt{\text{Var}(S)}$. Take

$$\pi_1 = \mathbb{E}[I_1] + \frac{c_\kappa \beta_1}{s_1} \sigma_S.$$

FOC for θ_1 :

$$\mathbb{E}\left[u_l \frac{\partial I_1}{\partial \theta_1}\right] - \lambda \left(\mathbb{E}\left[\frac{\partial I_1}{\partial \theta_1}\right] + c_\kappa \beta_1 \frac{\partial \sigma_S}{\partial \theta_1} \right) = 0.$$

Simplifying :

$$\frac{\partial \sigma_S}{\partial \theta_1} = \frac{1}{\sigma_S} \left(\underbrace{s_1^2 \text{Cov}(I_1, \frac{\partial I_1}{\partial \theta_1})}_{\text{own zone}} + \underbrace{s_1 s_2 \text{Cov}(I_2, \frac{\partial I_1}{\partial \theta_1})}_{\text{cross-zone}} \right).$$

SZ case: with $\sigma_{s_1 I_1} = s_1 \sigma(I_1)$,

$$\frac{\partial \sigma_{s_1 I_1}}{\partial \theta_1} = \frac{1}{\sigma_{s_1 I_1}} s_1^2 \text{Cov}(I_1, \frac{\partial I_1}{\partial \theta_1}), \quad (\text{no Cov}(I_2, \cdot) \text{ term}).$$

Multiple Zone Case

In the multi-zone case, the problems are coupled through the required capital

$$K = \text{CVaR}_{1-\epsilon} \left(\sum_z s_z I_z(\hat{\ell}_z) \right) - \mathbb{E} \left[\sum_z s_z I_z(\hat{\ell}_z) \right]$$

The first term will be affected by the correlation of losses across zones, so the optimal contracts will vary depending on this correlation. There are also additional decision variables k_z .

SZ vs MZ — what changes in the optimization problem?

$$\begin{aligned} \max_{a,b,K,\pi} \quad & \mathbb{E} \left[\sum_z s_z U(w_{0,z} + 1 - \ell_z - \pi_z + I_z(\theta_z)) \right] \\ \text{s.t.} \quad & \pi_z = \mathbb{E} \left[\overline{I_z(\theta_z)} \right] + \frac{c_\kappa}{s_z} k_z \end{aligned} \quad (7)$$

$$K = \text{CVaR}_{1-\epsilon} \left(\sum_z s_z \overline{I_z(\theta_z)} \right) - \mathbb{E} \left[\sum_z s_z \underline{I_z(\theta_z)} \right] \quad (8)$$

$$K = \sum_z k_z \quad (9)$$

$$\pi_z \leq \overline{\pi_z} \quad (10)$$

$$\overline{I_z(\theta_z)} = \max \left\{ 0, a_z \hat{\ell}_z(\theta_z) + b_z \right\}$$

$$\underline{I_z(\theta_z)} = \min \left\{ a_z \hat{\ell}_z(\theta_z) + b_z, 1 \right\}$$

SZ vs MZ — what changes in the optimization problem?

$$\max_{\{a_z, b_z\}_z, \{\pi_z\}_z, \{k_z\}_z, K} \mathbb{E} \left[\sum_z s_z U(w_{0,z} + 1 - \ell_z - \pi_z + \underline{l}_z(\theta_z)) \right] \quad (11)$$

$$\text{s.t.} \quad \pi_z = \mathbb{E} \left[\overline{l}_z(\theta_z) \right] + \frac{c_\kappa}{s_z} k_z \quad (\text{pricing with capital allocation})$$

$$K = \underbrace{\text{CVaR}_{1-\epsilon} \left(\sum_z s_z \overline{l}_z(\theta_z) \right) - \mathbb{E} \left[\sum_z s_z \underline{l}_z(\theta_z) \right]}_{\text{portfolio capital requirement}}$$

(portfolio risk couples zones)

$$K = \sum_z k_z \quad (\text{shared capital budget})$$

$$\pi_z \leq \overline{\pi}_z \quad (12)$$

$$\overline{l}_z(\theta_z) = \max \left\{ 0, a_z \hat{\ell}_z(\theta_z) + b_z \right\}, \quad (13)$$

$$\underline{l}_z(\theta_z) = \min \left\{ a_z \hat{\ell}_z(\theta_z) + b_z, 1 \right\}. \quad (14)$$

Multi-zone weakly dominates independent SZ (CVaR; detailed 1-slide sketch)

Setup. Per-zone contract l_z and per-unit premium

$$\pi_z = \mathbb{E}[l_z] + \frac{c_\kappa}{s_z} k_z, \quad K_z^{SZ} \equiv \text{CVaR}(l_z) - \mathbb{E}[l_z], \quad K^{SZ} \equiv \sum_z s_z K_z^{SZ}.$$

Fix SZ contracts in MZ. Take (a^{SZ}, b^{SZ}) from each SZ problem and fix them in P_2 so payouts l_z are unchanged.

Capital comparison (CVaR subadditivity).

$$K^{MZ} = \text{CVaR}\left(\sum_z s_z l_z\right) - \mathbb{E}\left[\sum_z s_z l_z\right] \leq \sum_z s_z (\text{CVaR}(l_z) - \mathbb{E}[l_z]) = K^{SZ}.$$

Let $\Delta K \equiv K^{SZ} - K^{MZ} \geq 0$.

Feasible capital shares exist. Since $K^{MZ} \leq K^{SZ}$, we can set $k_z^{MZ} = s_z K_z^{SZ} - \frac{\Delta K}{|Z|}$.

Premiums weakly decrease (cap respected).

$$\pi_z^{MZ} = \mathbb{E}[l_z] + \frac{c_\kappa}{s_z} k_z^{MZ} \leq \mathbb{E}[l_z] + \frac{c_\kappa}{s_z} s_z K_z^{SZ} = \pi_z^{SZ} \leq \bar{\pi}.$$

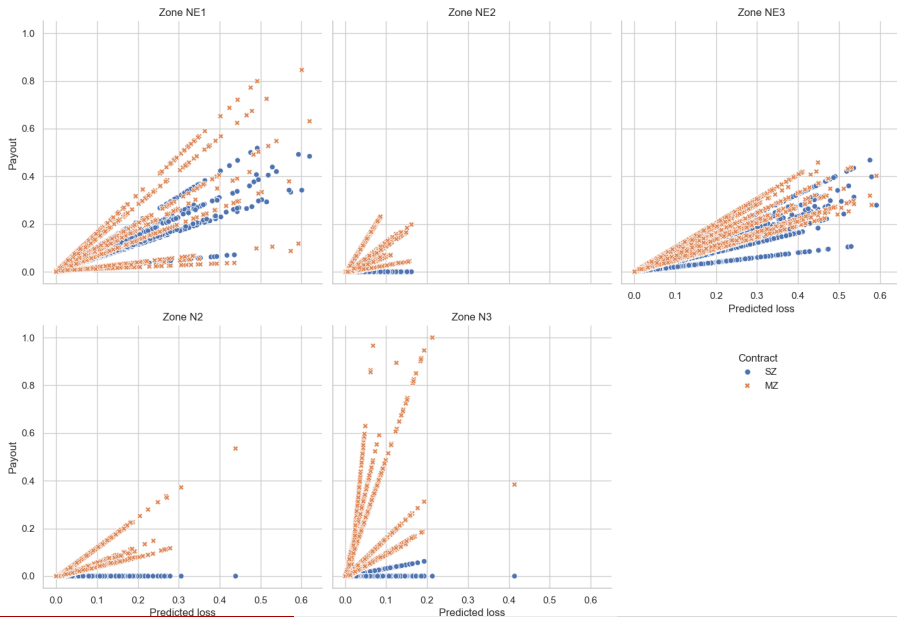
Conclusion. Payouts are identical and premiums are (weakly) lower, so utility weakly increases:

$$P_2^* \geq \sum_z P_1(z)^*.$$

MZ: joint design raises RIB 0.524 $\rightarrow$ 0.592 at realistic premiums

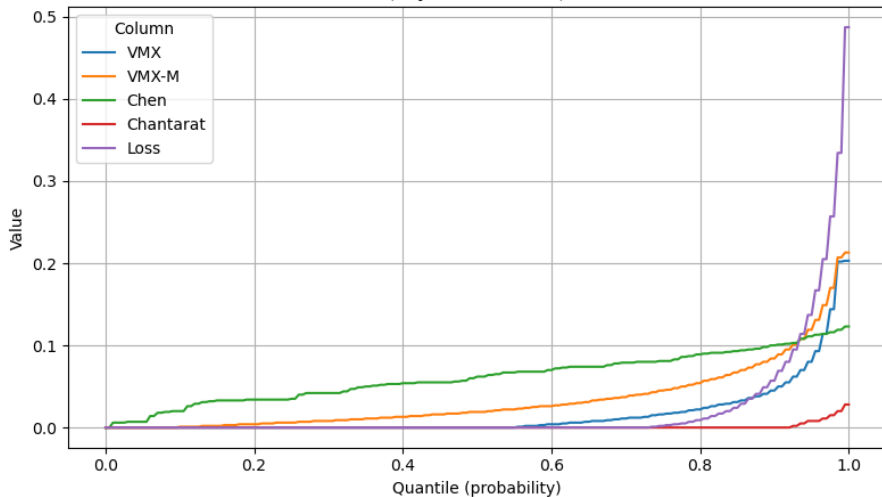
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Chantararat	-0.163	0.004	0.002	0.027
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MZ contracts tend to be steeper

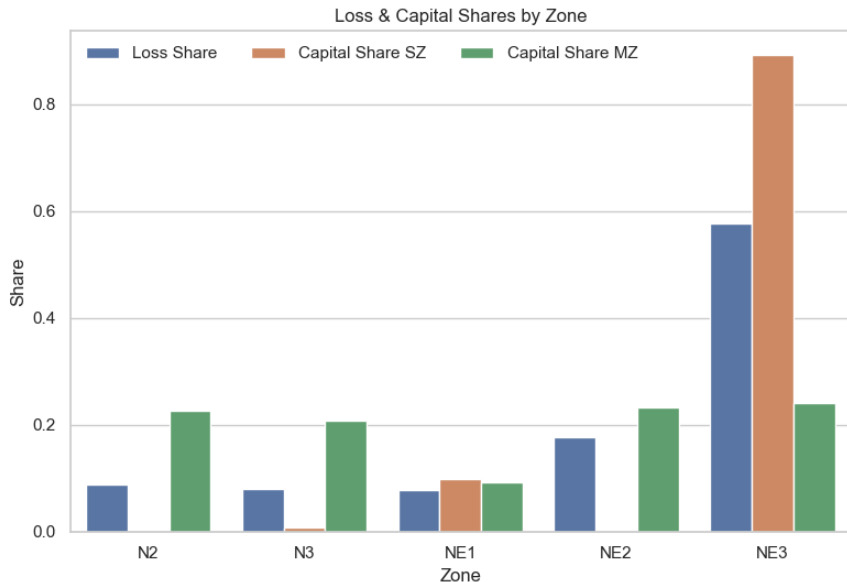


SZ vs MZ Payouts

Empirical Quantile Function
(Payouts and Loss)



Capital costs are reallocated to benefit high-loss zones



Insurance behavior: payout precision and loss recall

Definitions (normalized per plot i with weights w_i)

Let I_i be the (normalized) payout and L_i the realized loss.

$$\textbf{Payout Precision } P = \frac{\sum_i w_i \min\{I_i, L_i\}}{\sum_i w_i I_i} \quad (\text{share of payouts that were valid})$$

$$\textbf{Loss Recall } R = \frac{\sum_i w_i \min\{I_i, L_i\}}{\sum_i w_i L_i} \quad (\text{share of loss dollars that were covered})$$

- P penalizes over-payment; R penalizes under-payment when $L_i > 0$.
- Multi-zone (portfolio) design boosts R by using diversification and capital allocation to make larger payouts.

Utility gains/losses from insurance (relative to no insurance)

$$\Delta U = U(w_0 + 1 - \ell - \pi + I) - U(w_0 + 1 - \ell)$$

	Payout ($I > 0$)	No payout ($I = 0$)
Loss occurs	Gain — Good gain : loss is covered by payout; typically $\Delta U \geq 0$.	Loss — Bad loss : loss with no payout and premium; $\Delta U < 0$.
No loss	Gain : payout despite no loss; $\Delta U > 0$.	Loss : premium cost when no loss; $\Delta U < 0$.

Terminology: We call the first type of utility *gain* a **good gain** (loss with payout). We call the second type of utility *loss* a **bad loss** (loss without payout, plus premium).

Good gains and bad losses: welfare-focused view

Let $\Delta u_i \equiv u(w_i^{\text{ins}}) - u(w_i^0)$.

Metrics we report

$$\text{Good Gain} = \frac{\sum_{i:L_i>0} w_i (\Delta u_i)_+}{\sum_{i:\Delta u_i>0} w_i (\Delta u_i)_+}$$

share of utility gains that occur when there is a loss

$$\text{Bad Loss} = \frac{\sum_{i:\Delta u_i<0 \wedge L_i>0} w_i (\Delta u_i)_+}{\sum_{i:\Delta u_i<0} w_i (\Delta u_i)_+}$$

share of utility losses that occur when there is a loss

$$(x)_+ = \max\{x, 0\}$$

Interpretation

- **Higher** Good Gain $\Rightarrow$ insurance is behaving as intended.
- **Lower** Bad Loss $\Rightarrow$ less basis risk.

Insurance behavior results (VMX-M vs VMX and baselines)

Method	RIB	Payout Precision	Loss Recall	Good Gain	Bad Loss
VMX-M	0.597	0.184	0.211	0.390	0.334
VMX	0.524	0.175	0.120	0.335	0.372
Chantararat	-0.222	0.328	0.010	0.616	0.514
Chen	-1.924	0.173	0.298	0.327	0.440
Δ VMX-M – VMX	+0.075	+0.009	+0.091	+0.055	–0.038

- **Multi-zone (VMX-M)** raises welfare (+0.075 RIB) by *covering more true losses* (Recall +0.091) and *reducing under-protected loss states* (Bad Loss ↓ 0.038).
- Chantararat's high “precision” reflects very few payouts overall (low recall); VMX-M improves balance.

Summary of results (policy-facing)

- **Welfare:** RIB improves from 0.524 (VMX) to **0.597** (VMX-M).
- **Behavior:** VMX-M boosts *loss recall* substantially and reduces *bad loss* (basis risk).
- **Mechanism:** Portfolio capital lowers total required capital and reallocates it to benefit higher-risk zones; optimal contracts become slightly steeper where it is beneficial.

Conclusion & next steps

Takeaways.

- We develop a method that yields simple, deductible-like contracts with risk-adjusted pricing that improves welfare.
- **Multi-zone design** weakly dominates separate design and improves coverage.

Implications.

- Pricing on portfolio CVaR is more cost-effective.

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Idealized CVaR Model

- **Objective:** conditional value at risk of the farmers' loss net of insurance.
- **Constraint 1:** piecewise linear structure of the contract.
- **Constraint 2:** budget constraint.
- **Constraint 3:** definition of required capital.

$$\min_{a,b,\pi,K} \text{CVaR}_{1-\epsilon}(\ell - I(\theta))$$

$$\text{s.t. } I(\theta) = \min\{(a\hat{\ell}(\theta) + b)^+, P\} \quad (15)$$

$$\mathbb{E}[I(\theta)] + c_\kappa K \leq B \quad (16)$$

$$K = (\text{CVaR}_{1-\epsilon}(I(\theta)) - \mathbb{E}[I(\theta)]) \quad (17)$$

The problem is non-convex, so we need convex approximations

We use the following approximations of $I(\theta)$ to make the problem convex:

$$\begin{aligned}\overline{I(\theta)} &\triangleq \max \left\{ 0, a\hat{\ell}(\theta) + b \right\} \\ \underline{I(\theta)} &\triangleq \min \{ a\hat{\ell}(\theta) + b, K \}\end{aligned}$$

- Note that $\overline{I(\theta)} \geq I(\theta)$ and $\overline{I(\theta)}$ is convex. Conversely, $\underline{I(\theta)} \leq I(\theta)$ and $\underline{I(\theta)}$ is concave.
- We replace $I(\theta)$ with either $\overline{I(\theta)}$ or $\underline{I(\theta)}$ where necessary to obtain conservative and convex approximations.
- We also need approximations or proxies for $E[I(\theta)]$ in constraint . We use $\pi_{SQ} = E[I_{SQ}(\theta)]$, where I_{SQ} is the contract designed using the status quo method, as a proxy for $E[I(\theta)]$ in constraint .